

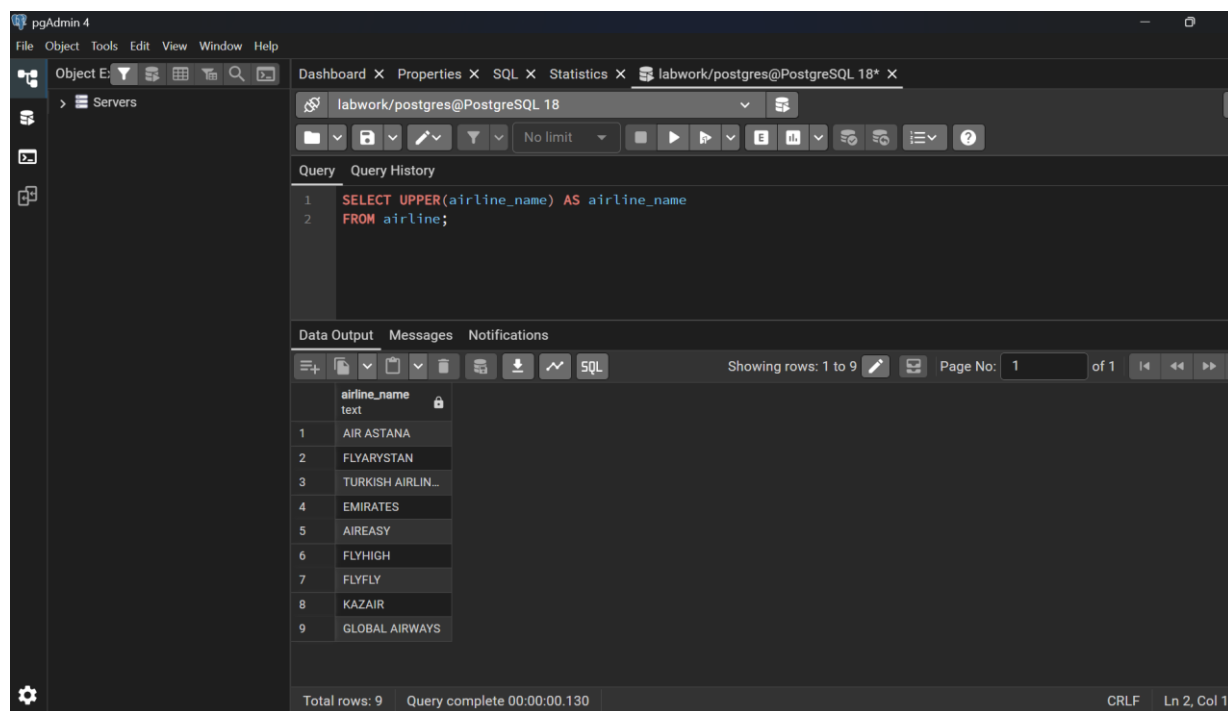
Laboratory work 4

We continue to work with the database from the previous laboratory works.

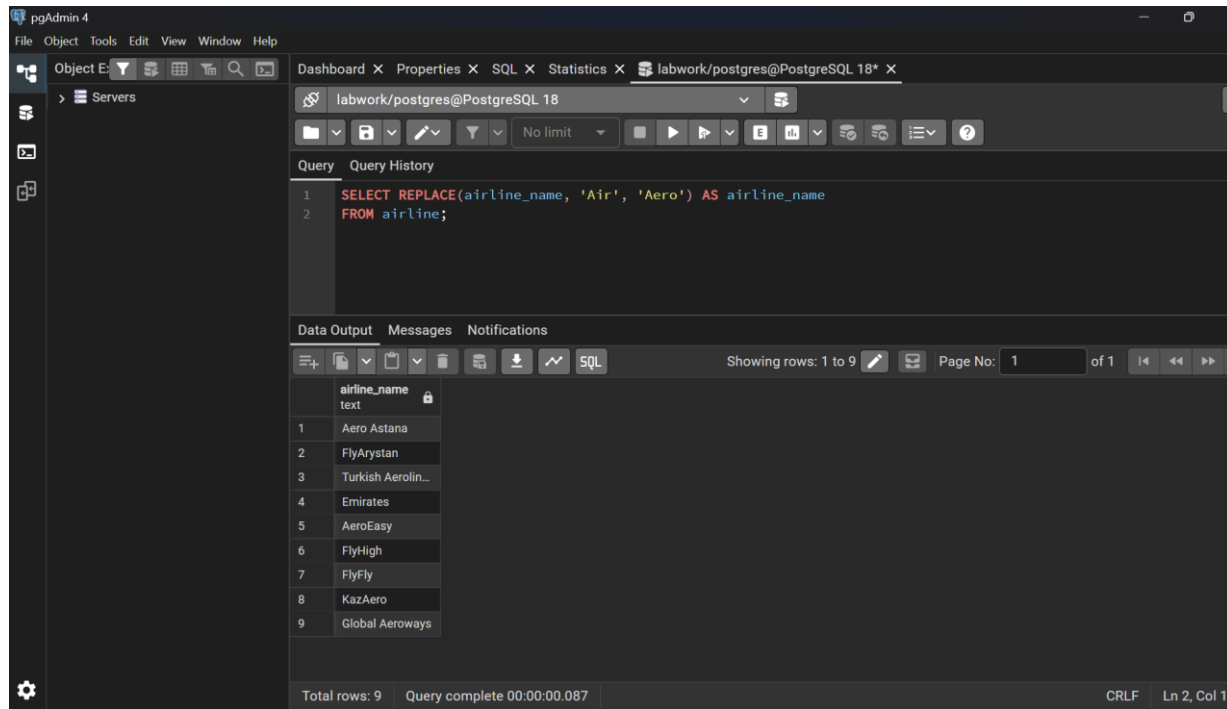
Take a full-page screenshot that covers the code and results of each task.

Tasks:

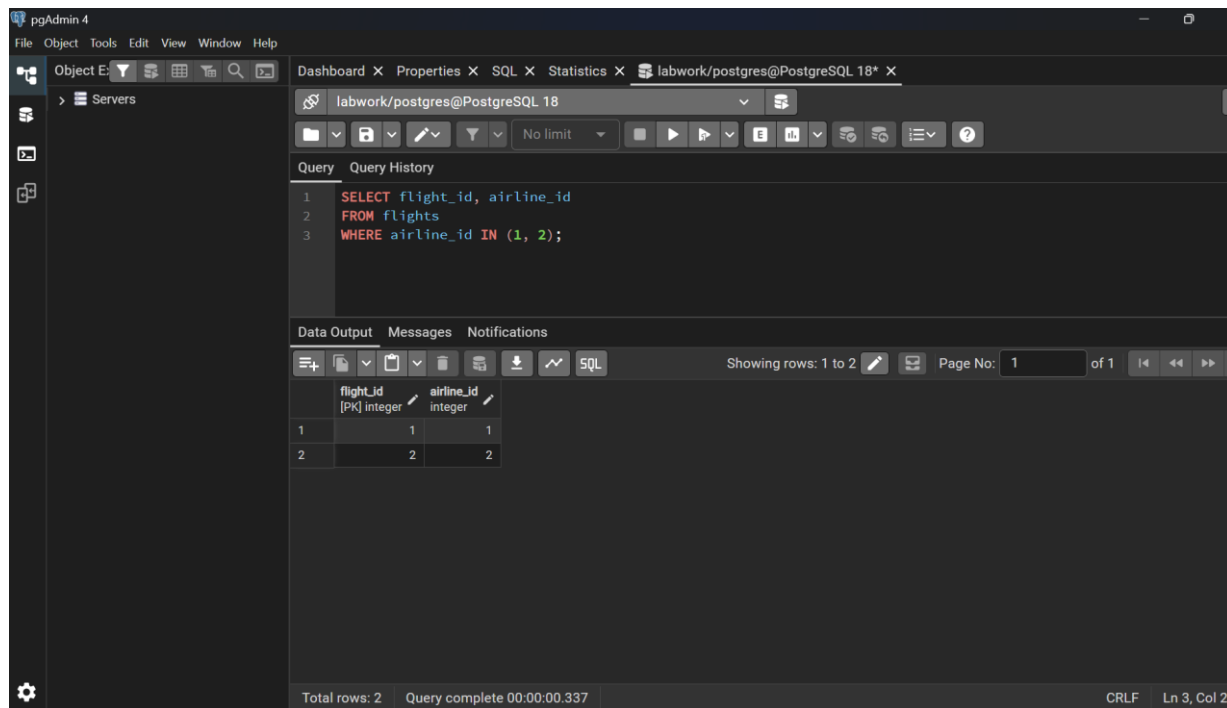
1. Retrieve all airline names in uppercase.



2. Replace any occurrence of the word "Air" in airline names with "Aero".



3. Find all flight numbers that coordinates with both airline 1 and airline 2.



4. Retrieve airports that contain the word "Reginal" and "Air" in their names.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer Servers

labwork/postgres@PostgreSQL 18*

labwork/postgres@PostgreSQL 18

Query Query History

```

1 SELECT *
2 FROM airport
3 WHERE airport_name ILIKE '%Reginal%'
4 OR airport_name ILIKE '%Air%';

```

Data Output Messages Notifications

Showing rows: 1 to 205 Page No: 1 of 1

	airport_id [PK] integer	airport_name character varying (50)	country character varying (50)	state character varying (50)	city character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	Almaty International Airp...	Kazakhstan	Almaty	Almaty	2026-09-23 00:04:38.332881	2026-09-23 00:04:38.332881
2	3	Istanbul Airport	Turkey	Istanbul	Istanbul	2026-09-23 00:04:38.332881	2026-09-23 00:04:38.332881
3	4	Dubai International Airport	UAE	Dubai	Dubai	2026-09-23 00:04:38.332881	2026-09-23 00:04:38.332881
4	5	Frankfurt Airport	Germany	Hesse	Frankfurt	2026-09-23 00:04:38.332881	2026-09-23 00:04:38.332881
5	101	Airport 1	Country 1	State 1	City 1	2026-09-30 10:39:29.829014	2026-09-30 10:39:29.829014
6	102	Airport 2	Country 2	State 2	City 2	2026-09-30 10:39:29.829014	2026-09-30 10:39:29.829014
7	103	Airport 3	Country 3	State 3	City 3	2026-09-30 10:39:29.829014	2026-09-30 10:39:29.829014
8	104	Airport 4	Country 4	State 4	City 4	2026-09-30 10:39:29.829014	2026-09-30 10:39:29.829014
9	105	Airport 5	Country 5	State 5	City 5	2026-09-30 10:39:29.829014	2026-09-30 10:39:29.829014
10	106	Airport 6	Country 6	State 6	City 6	2026-09-30 10:39:29.829014	2026-09-30 10:39:29.829014

Total rows: 205 Query complete 00:00:00.164 CRLF Ln 4, Col 3

- Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.

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File Object Tools Edit View Window Help

Object Explorer Servers

labwork/postgres@PostgreSQL 18*

labwork/postgres@PostgreSQL 18

Query Query History

```

1 SELECT first_name, last_name,
2 TO_CHAR(date_of_birth, 'Month DD, YYYY') AS birth_date
3 FROM passengers;

```

Data Output Messages Notifications

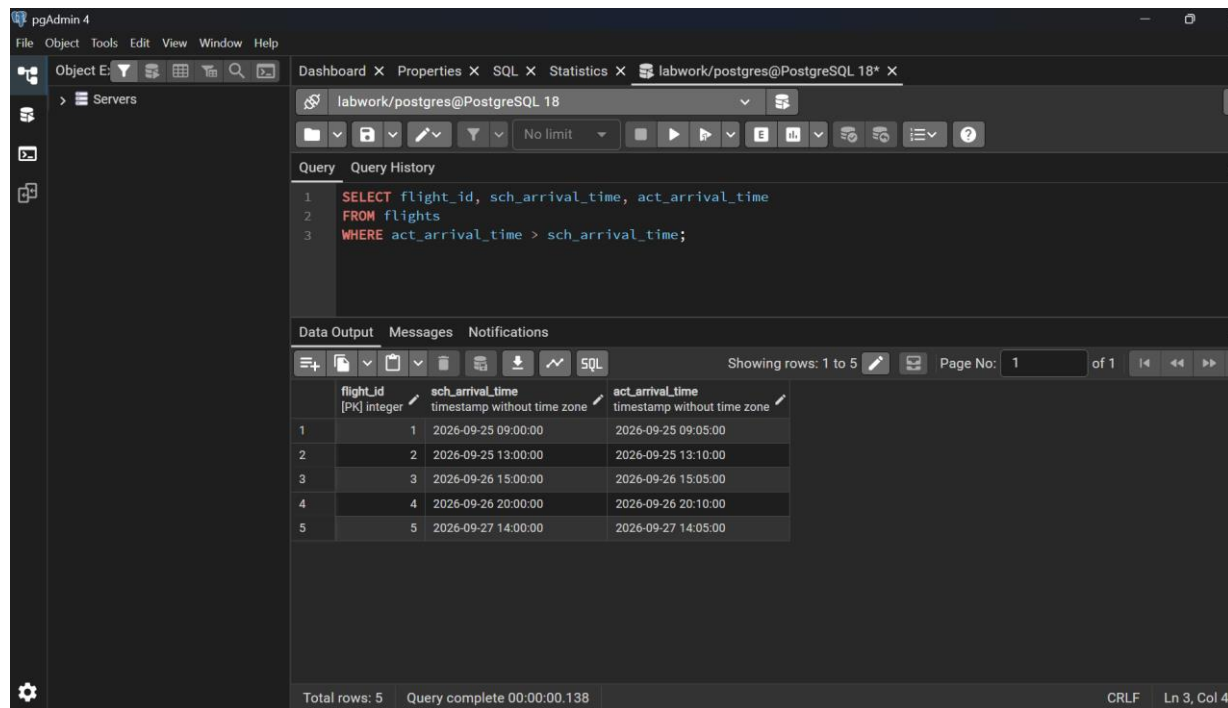
Showing rows: 1 to 5 Page No: 1 of 1

	first_name character varying (50)	last_name character varying (50)	birth_date text
1	Yeldos	Abdimalik	March 15, 2005
2	Ayan	Sarsenov	July 21, 2004
3	Dana	Bekova	January 10, 2005
4	Arman	Kaliyev	November 05, 20...
5	Aruzhan	Nurgaliyeva	September 18, 20...

Successfully run. Total query runtime: 160 msec. 5 rows affected.

Total rows: 5 Query complete 00:00:00.160 CRLF Ln 3, Col 1

- Find flight numbers that have been delayed based on the actual arrival time.



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File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```

1 SELECT flight_id, sch_arrival_time, act_arrival_time
2 FROM flights
3 WHERE act_arrival_time > sch_arrival_time;

```

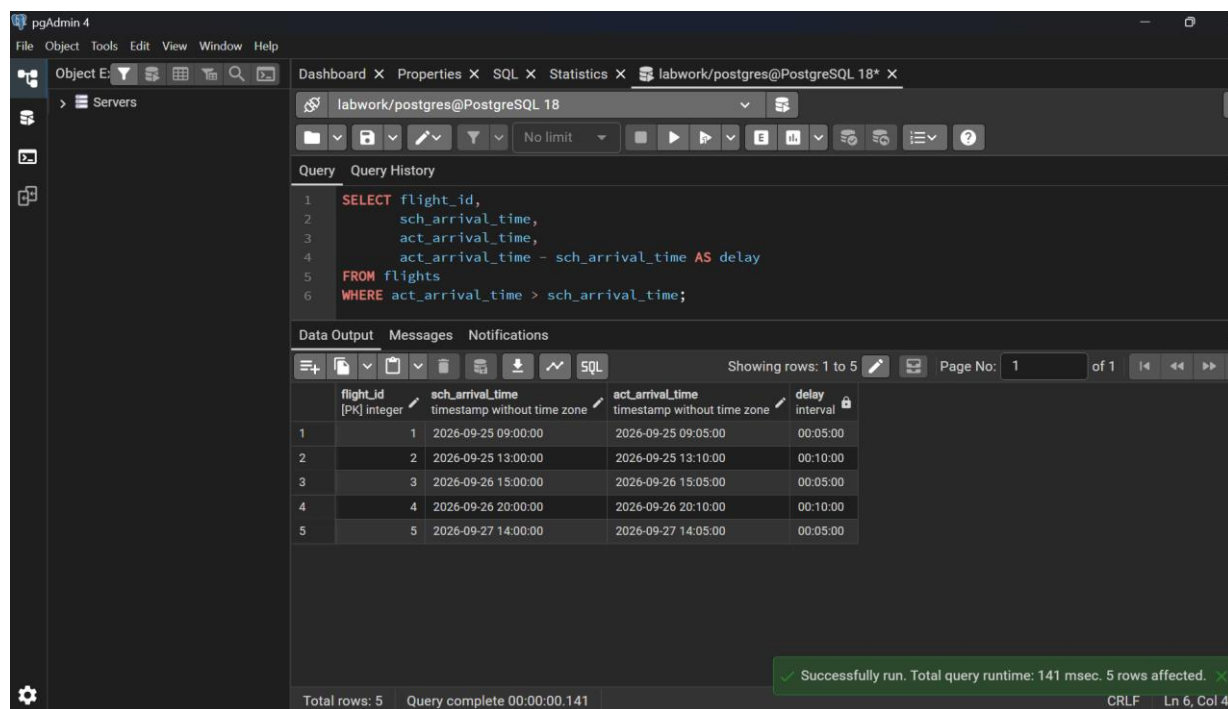
Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	flight_id [PK] integer	sch_arrival_time timestamp without time zone	act_arrival_time timestamp without time zone
1	1	2026-09-25 09:00:00	2026-09-25 09:05:00
2	2	2026-09-25 13:00:00	2026-09-25 13:10:00
3	3	2026-09-26 15:00:00	2026-09-26 15:05:00
4	4	2026-09-26 20:00:00	2026-09-26 20:10:00
5	5	2026-09-27 14:00:00	2026-09-27 14:05:00

Total rows: 5 Query complete 00:00:00.138 CRLF Ln 3, Col 4

- Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.



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File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```

1 SELECT flight_id,
2 sch_arrival_time,
3 act_arrival_time,
4 act_arrival_time - sch_arrival_time AS delay
5 FROM flights
6 WHERE act_arrival_time > sch_arrival_time;

```

Data Output Messages Notifications

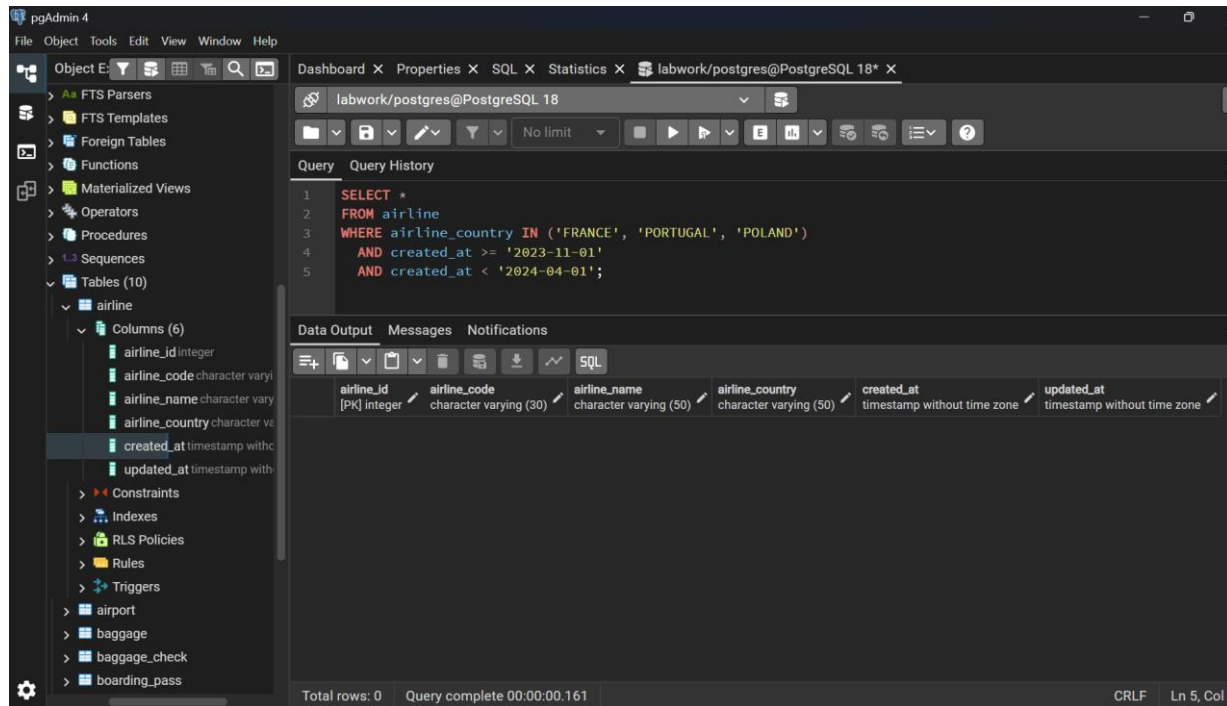
Showing rows: 1 to 5 Page No: 1 of 1

	flight_id [PK] integer	sch_arrival_time timestamp without time zone	act_arrival_time timestamp without time zone	delay interval
1	1	2026-09-25 09:00:00	2026-09-25 09:05:00	00:05:00
2	2	2026-09-25 13:00:00	2026-09-25 13:10:00	00:10:00
3	3	2026-09-26 15:00:00	2026-09-26 15:05:00	00:05:00
4	4	2026-09-26 20:00:00	2026-09-26 20:10:00	00:10:00
5	5	2026-09-27 14:00:00	2026-09-27 14:05:00	00:05:00

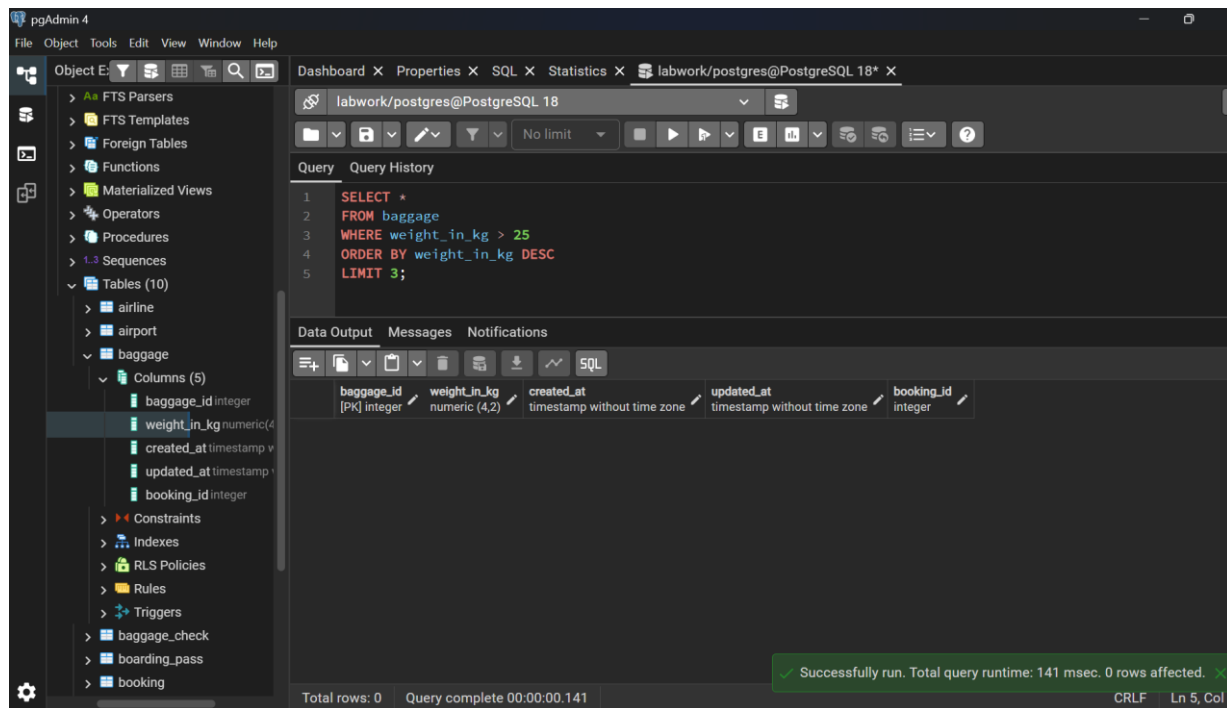
Successfully run. Total query runtime: 141 msec. 5 rows affected.

Total rows: 5 Query complete 00:00:00.141 CRLF Ln 6, Col 4

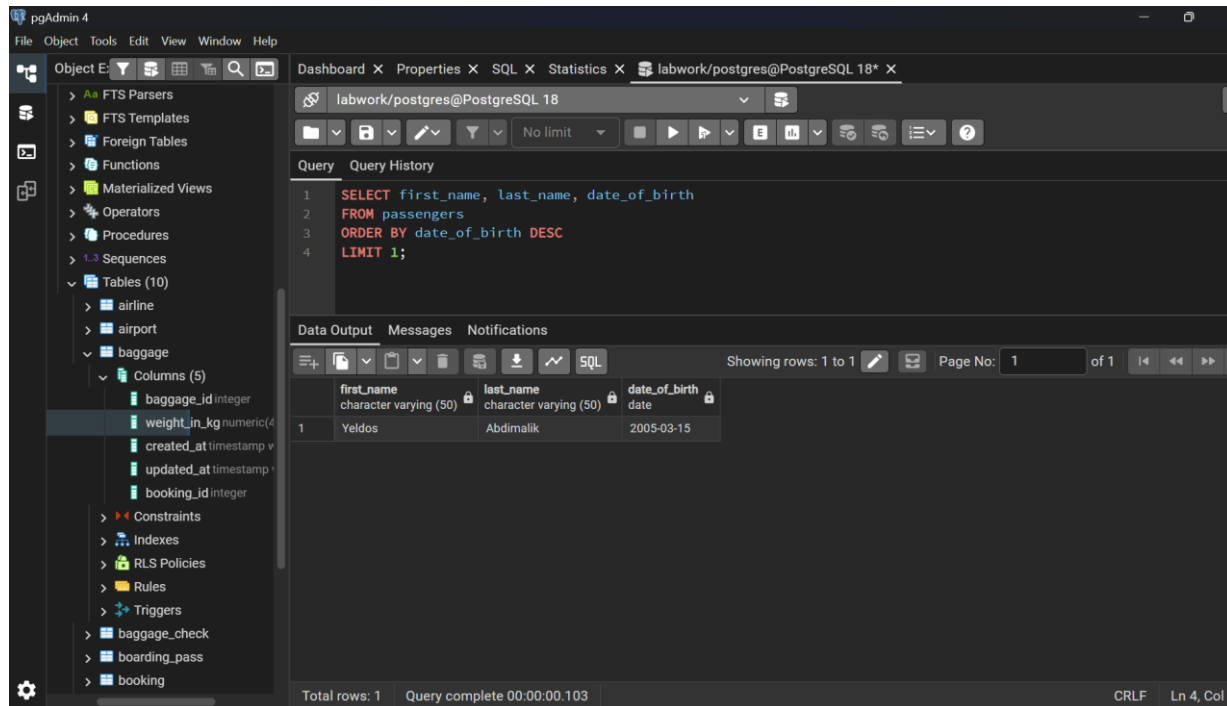
- Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.



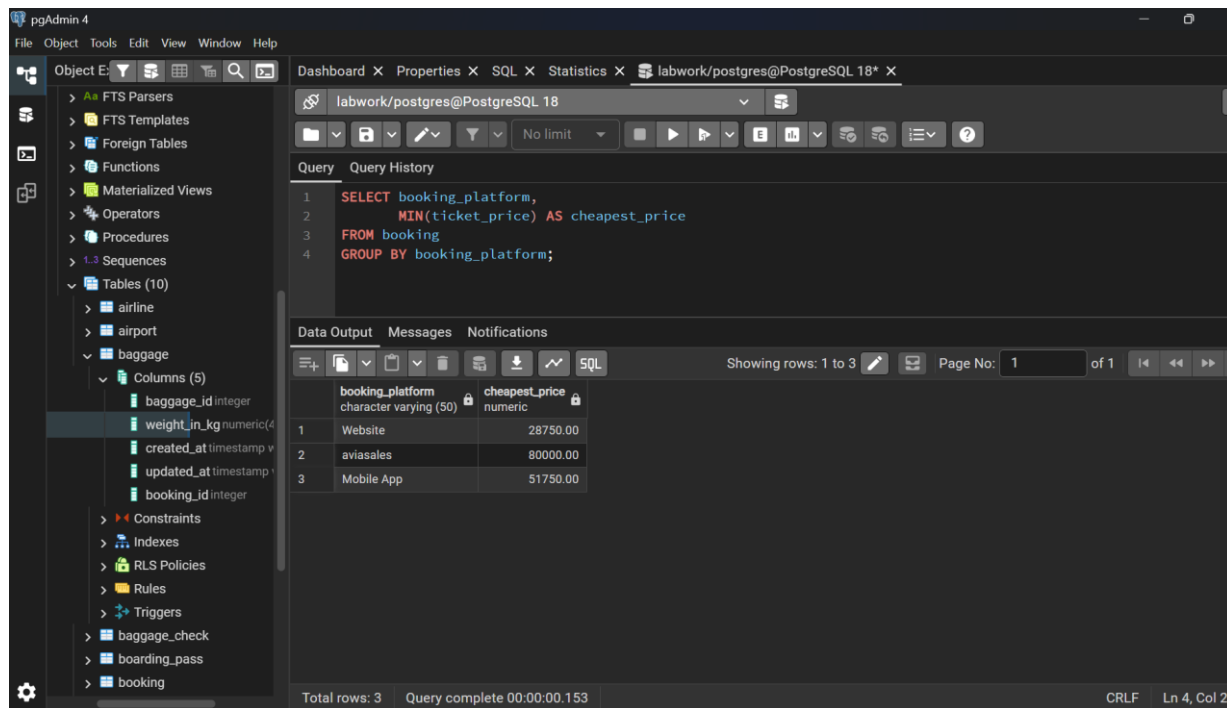
9. Find top3 overweighted baggage with more than 25kg.



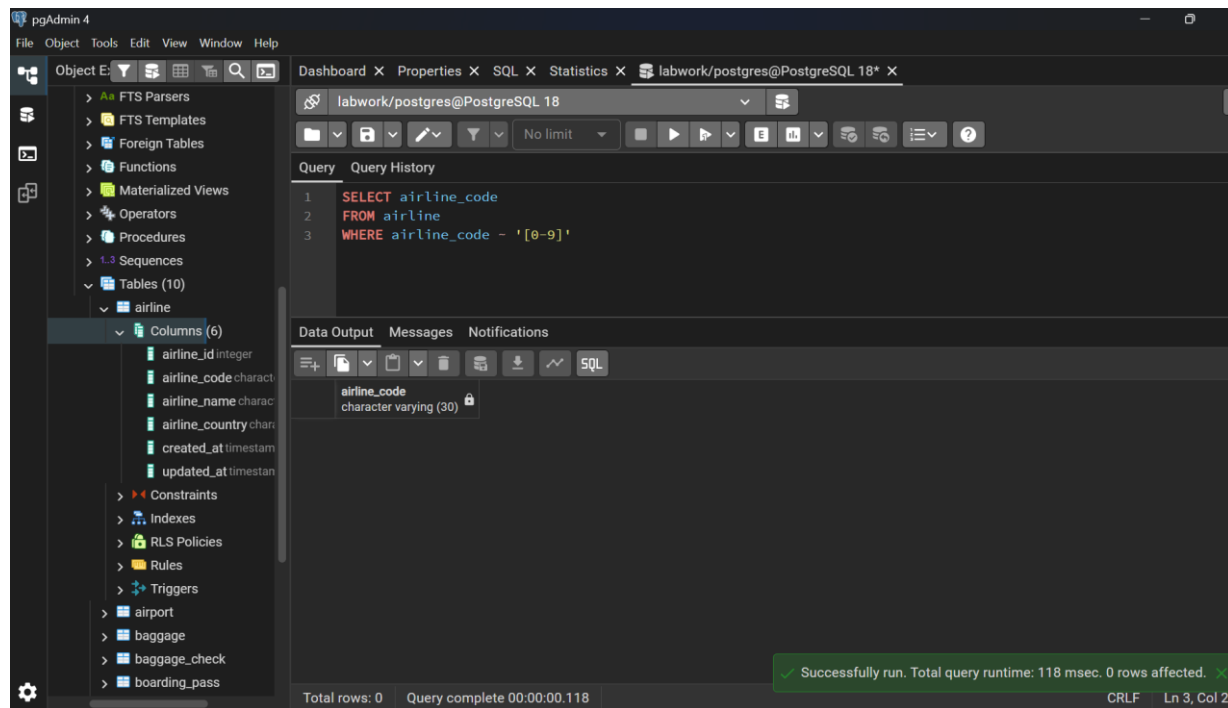
10. Find the youngest passengers' full name.



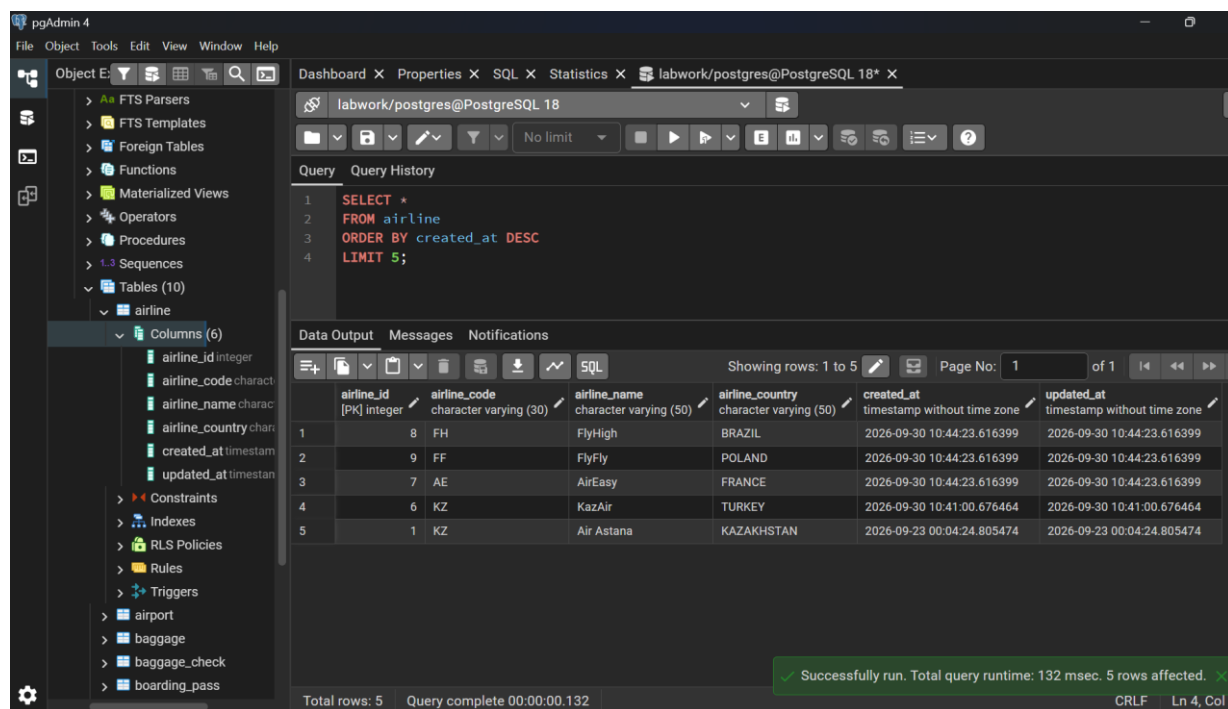
11. Find the cheapest booking price on each booking platform.



12. Return airlines whose airline_code contains a digit.



13. List the top5 most recently created airlines.



14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

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File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```

1 SELECT *
2 FROM baggage_check
3 WHERE booking_id BETWEEN 200 AND 300
4    OR check_result <> 'Checked';

```

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	1	1
2	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	2	2
3	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	3	3
4	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	4	4
5	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	5	5
6	Not checked	2026-09-30 12:01:58.372204	2026-09-30 12:01:58.372204	1	1

Total rows: 6 Query complete 00:00:00.176 CRLF Ln 4, Col 3

15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

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File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```

1 SELECT *
2 FROM baggage_check
3 WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
4    OR updated_at < created_at;

```

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	1	1
2	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	2	2
3	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	3	3
4	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	4	4
5	Passed	2026-09-23 00:08:07.09515	2026-09-23 00:08:07.09515	5	5
6	Not checked	2026-09-30 12:01:58.372204	2026-09-30 12:01:58.372204	1	1

Total rows: 6 Query complete 00:00:00.129 CRLF Ln 4, Col